

$$\begin{aligned} & -\frac{1}{168} g_1^4 y_d^{i112} L F_{2,2,-1}[m_1, m_d^{i12}] + \\ & -\frac{1}{18} g_1^2 y_d^{i112} \overline{y_u}^{i1r} L F_{2,1,0}[m_1, m_q^i] - \frac{1}{18} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[m_1, m_q^i] + \\ & -\frac{1}{18} g_1^2 y_d^{i12} \overline{y_u}^{i1r} a_u^{i1r} L F_{4,1,-2}[m_1, m_q^i] + \frac{1}{864} g_1^4 y_d^{i112} (g_1^2 - 3 g_2^2) L F_{2,-2,-1}[m_1, m_q^{i11}] + \\ & -\frac{1}{9} g_1^2 y_d^{i12} \overline{y_u}^{i1r} L F_{2,1,0}[m_1, m_q^i] - \frac{8}{9} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[m_1, m_q^i] + \\ & -\frac{1}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} L F_{4,1,-2}[m_1, m_q^i] + \frac{1}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{2,1,0}[m_2, m_q^i] - \\ & -\frac{1}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[m_2, m_q^i] + \frac{1}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{4,1,-2}[m_2, m_q^i] + \\ & -\frac{1}{3} g_1^2 y_d^{i12} (g_1^2 + g_2^2) L F_{2,2,-1}[m_2, m_q^{i1}] + \frac{1}{6} g_1^2 y_d^{i12} \overline{y_u}^{i1r} L F_{2,2,-1}[m_3, m_d^{i12}] + \\ & -\frac{1}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} L F_{2,1,0}[m_3, m_q^i] - \frac{8}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[m_3, m_q^i] + \\ & -\frac{1}{18} g_1^2 y_d^{i12} \overline{y_u}^{i1r} L F_{4,1,-2}[m_3, m_q^i] + \frac{1}{18} g_1^2 y_d^{i12} (g_1^2 - 3 g_2^2) L F_{2,2,-1}[m_3, m_q^{i1}] + \\ & -\frac{8}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} L F_{2,1,0}[m_3, m_q^i] - \frac{8}{3} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[m_3, m_q^i] + \\ & -\frac{1}{24} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_d^{i12} y_d^{i1r} L F_{2,2,-1}[m_d^r, \tilde{\mu}] - \\ & -\frac{2}{3} g_1^2 y_d^{i12} L F_{2,1,0}[m_d^{i2}, m_1] - \frac{2}{3} g_1^2 g_2^2 y_d^{i12} L F_{2,1,0}[m_d^{i2}, m_3] + \\ & -\frac{1}{2} g_1^2 y_d^{i12} \overline{a_u}^{i1r} L F_{2,2,0}[m_d^r, \tilde{\mu}] - \frac{1}{2} g_2^2 y_d^{i12} \overline{a_u}^{i1r} a_u^{i1r} L F_{3,1,0}[m_d^r, m_e^r] - \\ & -\frac{1}{8} y_d^{i12} \overline{a_u}^{i1r} a_u^{i1r} L F_{3,2,-1}[m_d^r, m_e^r] - \frac{3}{8} y_d^{i12} \overline{a_u}^{i1r} a_u^{i1r} (g_1^2 - 3 g_2^2) L F_{4,1,-1}[m_d^r, m_e^r] + \\ & -\frac{2}{3} a_u^{i1r} a_e^{i1r} (g_1^2 - g_2^2) L F_{5,1,-2}[m_d^r, m_e^r] + \\ & -\frac{1}{4} g_1^2 y_d^{i12} \overline{a_u}^{i1r} a_u^{i1r} L F_{3,2,-1}[m_d^r, m_e^r] - \frac{1}{4} g_1^2 y_d^{i12} \overline{a_u}^{i1r} a_u^{i1r} L F_{3,2,-1}[m_d^r, m_e^r] + \\ & -\frac{1}{4} g_1^2 y_d^{i12} \overline{a_u}^{i1r} a_u^{i1r} (7 g_1^2 - 9 g_2^2) L F_{4,1,-1}[m_d^r, m_e^r] + \\ & -\frac{1}{4} g_1^2 y_d^{i12} \overline{a_u}^{i1r} a_u^{i1r} (g_1^2 - g_2^2) L F_{5,1,-2}[m_d^r, m_e^r] + \\ & -\frac{1}{12} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} (g_1^2 + 3 g_2^2) L F_{2,2,0}[m_d^r, m_e^r] - \frac{1}{12} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{2,1,0}[m_d^r, \tilde{\mu}] + \\ & -\frac{1}{432} g_1^2 y_d^{i12} (g_1^2 - 3 g_2^2) L F_{2,1,0}[m_q^{i1}, m_1] - \\ & -\frac{1}{9} g_1^2 y_d^{i12} (g_1^2 - g_2^2) L F_{2,2,-1}[m_q^{i1}, m_3] - \\ & -\frac{1}{8} \tilde{\mu}^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} (g_1^2 + 3 g_2^2) L F_{3,1,0}[m_q^i, m_q^i] - \frac{1}{8} \tilde{\mu}^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} (g_1^2 + 3 g_2^2) \\ & -\frac{3}{2} \tilde{\mu}^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} (g_1^2 - 3 g_2^2) L F_{4,1,-1}[m_q^i, m_q^i] + \\ & -\frac{1}{6} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} (g_1^2 - g_2^2) L F_{5,1,-2}[m_q^i, m_q^i] + \frac{1}{6} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{2,1,0}[m_q^i, \tilde{\mu}] - \\ & -\frac{1}{2} \overline{y_u}^{i1r} y_d^{i12} y_u^{i1r} L F_{2,2,-1}[m_q^i, \tilde{\mu}] + \frac{3}{2} \overline{y_u}^{i1r} y_d^{i12} y_u^{i1r} L F_{2,1,0}[m_q^i, m_d^r] - \\ & -\frac{1}{2} g_1^2 y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[m_q^i, m_d^r] + \frac{1}{2} g_1^2 y_d^{i12} (-g_1^2 + g_2^2) L F_{3,1,-1}[\tilde{\mu}, m_1] - \\ & -\frac{3}{8} g_1^4 m_1^2 y_d^{i12} L F_{3,2,-1}[\tilde{\mu}, m_1] + \\ & -\frac{1}{2} g_1^2 y_d^{i12} L F_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^4 y_d^{i12} L F_{4,2,-3}[\tilde{\mu}, m_1] + \\ & -\frac{1}{2} g_1^2 y_d^{i12} L F_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^2 y_d^{i12} (-g_1^2 + g_2^2) L F_{5,1,-3}[\tilde{\mu}, m_1] + \\ & -\frac{1}{15} g_1^2 y_d^{i12} L F_{3,2,-2}[\tilde{\mu}, m_2] - \\ & -\frac{1}{2} g_1^2 y_d^{i12} L F_{3,2,-1}[\tilde{\mu}, m_2] + 3 g_1^2 y_d^{i12} (g_1^2 - g_2^2) L F_{4,1,-2}[\tilde{\mu}, m_2] + \\ & -\frac{1}{4} g_1^2 y_d^{i12} L F_{4,2,-3}[\tilde{\mu}, m_2] + \frac{1}{4} g_1^2 y_d^{i12} (m_2^2 + 2 \tilde{\mu}^2) L F_{4,2,-2}[\tilde{\mu}, m_2] + \\ & -\frac{1}{2} \overline{y_u}^{i1r} y_d^{i12} L F_{5,1,-3}[\tilde{\mu}, m_2] + \frac{1}{2} \overline{y_u}^{i1r} y_d^{i12} y_d^{i1r} \overline{y_u}^{i1r} L F_{2,1,0}[\tilde{\mu}, m_d^r] - \\ & -\frac{1}{2} y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[\tilde{\mu}, m_d^r] + \frac{1}{2} y_d^{i12} \overline{y_u}^{i1r} y_d^{i1r} y_u^{i1r} L F_{4,1,-2}[\tilde{\mu}, m_d^r] - \\ & -\frac{1}{2} y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{3,1,-1}[\tilde{\mu}, m_d^r] + \frac{1}{2} y_d^{i12} \overline{y_u}^{i1r} y_u^{i1r} L F_{4,1,-2}[\tilde{\mu}, m_d^r] + \\ & -\frac{1}{2} y_d^{i12} \overline{y_u}^{i1r} a_d^{i1r} L F_{2,1,1,0}[m_1, m_d^{i2}, m_q^i] - \\ & -\frac{1}{36} m_1 g_1^2 y_d^{i12} \overline{y_u}^{i1r} a_d^{i1r} L F_{2,2,1,-1}[m_1, m_d^{i2}, m_q^i] + \\ & -\frac{1}{144} m_1 g_1^2 a_d^{i1r} L F_{2,2,1,-1}[m_1, m_d^{i2}, m_q^i] + \\ & -\frac{1}{9} g_1^2 y_d^{i12} \overline{y_u}^{i1r} L F_{2,1,1,-1}[m_1, m_d^{i2}, m_q^i] - \\ & -\frac{1}{18} g_1^4 m_1^2 y_d^{i12} L F_{2,1,1,0}[m_1, m_d^{i2}, \tilde{\mu}] + \\ & -\frac{1}{18} g_1^4 m_1^2 y_d^{i12} L F_{3,1,1,-2}[m_1, m_d^{i2}, \tilde{\mu}] + \frac{1}{18} g_1^4 m_1^2 y_d^{i12} L F_{2,1,1,-1}[m_1, m_d^{i2}, \tilde{\mu}] + \\ & -\frac{1}{2} y_d^{i12} \overline{y_u}^{i1r} a_d^{i1r} L F_{2,2,1,-1}[m_1, m_d^{i2}, m_q^i] - \frac{1}{9} m_1 g_1^2 y_d^{i12} y_u^{i1r} \overline{a_u}^{i1r} \\ & -\frac{1}{9} m_1 g_1^2 y_d^{i12} y_u^{i1r} \overline{a_u}^{i1r} L F_{3,1,1,-1}[m_1, m_q^i, m_u^r] - \\ & -\frac{1}{12} g_1^2 y_d^{i1$$